Viennese physicist Erwin Schrödinger published a competing theory of quantum physics: wave mechanics.

Schrödinger had come up with wave mechanics while shacked up with his mistress in a resort in the Swiss Alps in December 1925. His theory was written in the relatively simple mathematical language of waves, with smoothly changing wave functions governed by the Schrödinger equation (as we saw in Chapter 1). Heisenberg was worried that Schrödinger's accomplishment would eclipse his own, and rightly so. The abstruse mathematics of Heisenberg's matrix mechanics was unfamiliar to most physicists at the time, and it had no obvious picture of the world to go along with it. Schrödinger's theory, meanwhile, used familiar mathematics with simple physical ideas; it was easy to handle and easy to think about. Schrödinger boasted that with his theory, physicists didn't have to "suppress intuition and to operate only with abstract concepts such as transition probabilities, energy levels, and the like." Much of the physics community agreed, even Heisenberg's erstwhile allies. Arnold Sommerfeld, Heisenberg's PhD adviser, said, "Although the truth of matrix mechanics is indubitable, its handling is extremely intricate and frighteningly abstract. Schrödinger has now come to our rescue." Born described Schrödinger's wave mechanics as "the deepest form of the quantum laws." Pauli, meanwhile, used Schrödinger's theory to do what he had been unable to accomplish with matrix mechanics alone—he managed to derive the brightness of the spectral lines in hydrogen, solving a problem that had been outstanding for more than seventy years.

